IntentCloud: Three-Tier System Architecture & Communication Dataflow

On-Device Neural Document Search · Local Hybrid Embeddings · Calibrated Gate · Zero-Trust Transport

TIER 1: PRESENTATION & USER INTERACTION TIER [Next.js 14 · React 19 · TypeScript · Tailwind CSS]

Natural Language Query UI

- Real-time debounced search bar
- Dynamic category filter pills
- Client query caching & history
- Low-latency reactive state

Drag-and-Drop Ingestion

- Multi-format drop zone (PDF/DOCX/TXT)
- Client-side format & size guard
- Asynchronous batch upload tracker
- Live indexing progress telemetry

Attribution & Citations

- Confidence score visual badges
- Matched sentence verbatim excerpts
- Multi-document comparative view
- Low-confidence warning alerts

Zero-Trust File Delivery

- Authentic byte-stream download
- In-browser instant PDF preview
- Remote access via TLS reverse proxy
- Cryptographic SHA integrity check

⟵ REST API / JSON Payloads & Byte Streams (HTTP/2 localhost:8000 / Cloudflare Zero-Trust Tunnel) ⟶

TIER 2: COGNITIVE PROCESSING & SERVICE ORCHESTRATION TIER [FastAPI · Asynchronous ASGI · PyTorch ML Runtime]

Ingestion & Dual Vector Encoding

- PyMuPDF & python-docx native stream parsing
- Tesseract OCR automatic fallback (<100 chars)
- Sliding-window chunker (C=1200, O=200 chars)
- Dense: all-MiniLM-L6-v2 (384-dim embeddings)
- Sparse: MurmurHash3 feature hashing (D=10^6)
- SHA-256 fingerprinting & cosine deduplication

Intent Analysis & Hybrid Search

- Local Ollama runtime (llama3.2:1b, 4-bit quant)
- Query intent classification & synonym expansion
- Parallel dense semantic search (Cosine top-N=20)
- Parallel sparse lexical search (Dot product top-N=20)
- Reciprocal Rank Fusion (RRF, constant k=60)
- Candidate deduplication & pool pruning (N <= 25)

Cross-Attention Reranking & Gating

- Cross-Encoder: ms-marco-MiniLM-L-6-v2
- Full token-level cross-attention score s(q, c)
- Calibrated Confidence Gate threshold (tau = 0.40)
- Out-of-Distribution (OOD) active abstention
- Sentence-level provenance & attribution span
- Performance telemetry (P95 latency tracking)

⟵ In-Memory C ABI / Direct NVMe File I/O Streams (Air-Gapped, Zero Network Latency) ⟶

TIER 3: EMBEDDED MEMORY & LOCAL STORAGE TIER [Local NVMe Storage · Complete Air-Gap Sovereignty]

Embedded Vector Database

- Embedded Qdrant instance (path: ./qdrant_storage/)
- SQLite transactional store with Write-Ahead Logging (WAL)
- HNSW index for dense vectors + sparse inverted index
- Zero external daemon required; runs entirely in-process

Local Document Vault

- Air-gapped file repository (path: ./uploads/)
- Preserves exact byte structure of source documents
- Cryptographic SHA-256 integrity validation
- Tamper-proof local on-premise storage

Document Registry & Metadata

- Lightweight document catalog (path: metadata.json)
- Track original filenames, mime-types, and timestamps
- Document versioning & update lineage history
- Rapid cold-start index restoration on daemon boot